

Mathematical Foundations of Political Methodology

Robert Gulotty

University of Chicago

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Summarizing Random Variables

- Covariance
- Multivariate Distributions
- Law of Iterated Expectations (LIE)
- Law of Total Variance

Conditional Probability Distribution Function

Definition

Suppose X and Y are continuous random variables with joint pdf $f(x, y)$. Then define the *conditional probability function* $f(x|y)$ as

$$f(x|y) = \frac{f(x, y)}{f_Y(y)}$$

Covariance

Definition

For jointly continuous random variables X and Y define, the covariance of X and Y as,

$$\begin{aligned}\text{cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY - E[X]Y - E[Y]X + E[X]E[Y]] \\ &= E[XY] - 2E[X]E[Y] + E[E[X]E[Y]] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

Define the correlation of X and Y as,

$$\text{cor}(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Some Observations

Variance is the covariance of a random variable with itself.

$$\begin{aligned}\text{cov}(X, X) &= E[XX] - E[X]E[X] \\ &= E[X^2] - E[X]^2\end{aligned}$$

Correlation measures the linear relationship between two random variables

Suppose $X = Y$

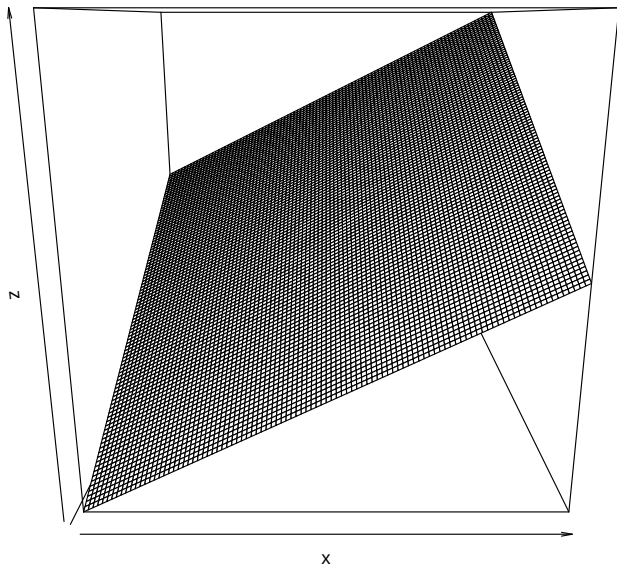
$$\begin{aligned}\text{cor}(X, Y) &= \frac{\text{cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \\ &= \frac{\text{Var}(X)}{\text{Var}(X)} \\ &= 1\end{aligned}$$

Suppose $X = -Y$

$$\begin{aligned}\text{cor}(X, Y) &= \frac{\text{cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \\ &= \frac{-\text{Var}(X)}{\text{Var}(X)}\end{aligned}$$

Plot pdf $f(x, y) = x + y$

```
x <- seq(0, 1, 0.01)
y <- seq(0, 1, 0.01)
z <- outer(x, y, function(x, y){ x + y })
persp(x, y, z)
```



Example: $X + Y$

Suppose X and Y have pdf $x + y$ for $x, y \in [0, 1]$.

$\text{Cov}(X, Y)$

$$\begin{aligned}E[XY] &= \int_0^1 \int_0^1 xy(x + y) dx dy \\&= \int_0^1 \int_0^1 (x^2 y + y^2 x) dx dy \\&= \int_0^1 \left(\frac{y}{3} + \frac{y^2}{2} \right) dy \\&= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}\end{aligned}$$

$$\begin{aligned}E[X] &= \int_0^1 \int_0^1 x(x + y) dx dy \\&= \frac{7}{12}\end{aligned}$$

$$E[Y] = \int_0^1 \int_0^1 y(x + y) dx dy$$

Example: $X + Y$

$$\begin{aligned}\text{Cov}(X, Y) &= E[XY] - E[X]E[Y] \\ &= \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}\end{aligned}$$

$$\begin{aligned}\text{Cor}(X, Y) &= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \\ &= \frac{-\frac{1}{144}}{\frac{11}{144}} \\ &= \frac{-1}{11}\end{aligned}$$

Conditional Expectation

$$\mathbb{E}(Y|X) = \int y \cdot f_{Y|X}(y|x) dy$$

$\mathbb{E}(Y|X)$ is the *best* way to predict Y given X .

Example of Joint Density

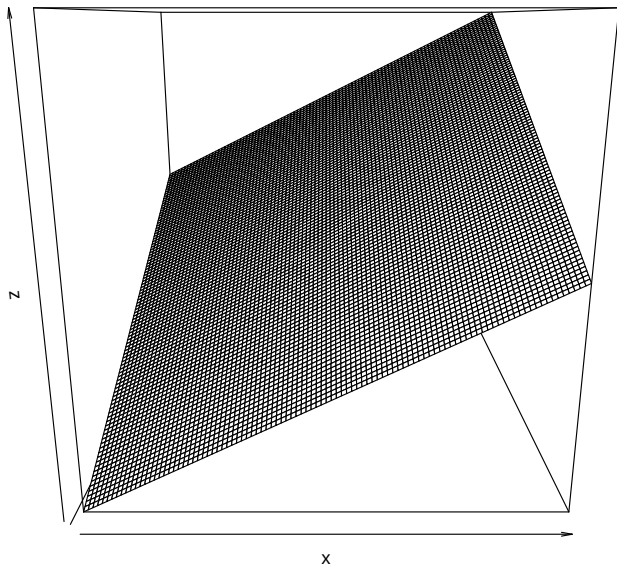
$$f_{X,Y}(x,y) = x + y, \quad \text{for } x, y \in [0,1]$$

the marginal densities $f_X(x)$ and $f_Y(y)$ are

$$f_X(x) = x + \frac{1}{2}$$

$$f_Y(y) = \frac{1}{2} + y$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{x + y}{\frac{1}{2} + y}$$



Example of Conditional Expectation

$$\begin{aligned} E(Y|X) &= \int_0^1 y f_{Y|X}(y|x) dy \\ &= \int_0^1 y \left(\frac{x+y}{\frac{1}{2}+x} \right) dy \\ &= \frac{1}{x+\frac{1}{2}} \int_0^1 y(x+y) dy \\ &= \frac{1}{x+\frac{1}{2}} \left(\frac{y^2 x}{2} + \frac{y^3}{3} \right) \Big|_0^1 \\ &= \frac{2+3x}{3+6x} \end{aligned}$$

Best Linear Predictor

Suppose we want a predictor $E(Y|X)$ that is linear:

$$f(X) = a + bX$$

Where a and b are set to minimize

$$E[(Y - (a + bX))^2]$$

Best Linear Predictor

$$E[(Y - (a + bX))^2]$$

Take partial derivative with respect to a .

$$\partial E[(Y - (a + bX))^2] / \partial a = E[\partial(Y - (a + bX))^2 / \partial a]$$

$$= -2E[(Y - (a + bX))]$$

setting equal to 0:

$$0 = -2E[(Y) - (a + bX)]$$

$$0 = -2E[Y] + 2E(a + bX)]$$

$$E[Y] = a + bE(X)$$

$$E[Y] - bE(X) = a$$

Best Linear Predictor

$$E[(Y - (a + bX))^2]$$

Take partial derivative with respect to b .

$$\partial E[(Y - (a + bX))^2] / \partial b = E[\partial(Y - (a + bX))^2 / \partial b]$$

$$= 2E[(Y - (a + bX))(-X)]$$

setting equal to 0:

$$0 = -2E[(YX) - (a + bX)(X)]$$

$$0 = -2E[YX] + 2E(a + bX)(X)]$$

Best Linear Predictor (continued)

$$E[YX] = aE(X) + bE(X^2)]$$

$$E[YX] = [E[Y] - bE(X)]E(X) + bE(X^2)]$$

$$E[YX] = E[Y]E(X) - bE(X)E(X) + bE(X^2)]$$

$$E[YX] - E[Y]E(X) = b[E(X^2) - E[X]^2]$$

$$b = \frac{E[YX] - E[Y]E(X)}{[E(X^2) - E[X]^2]} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

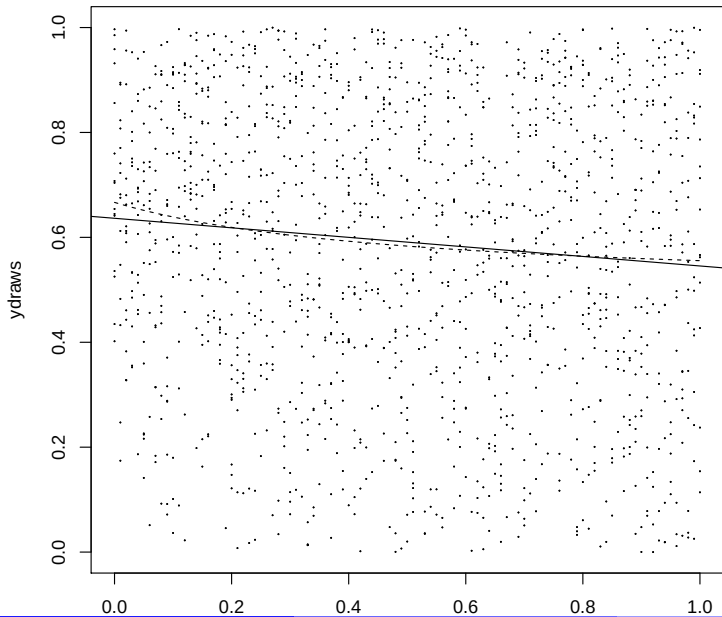
Best Linear Predictor

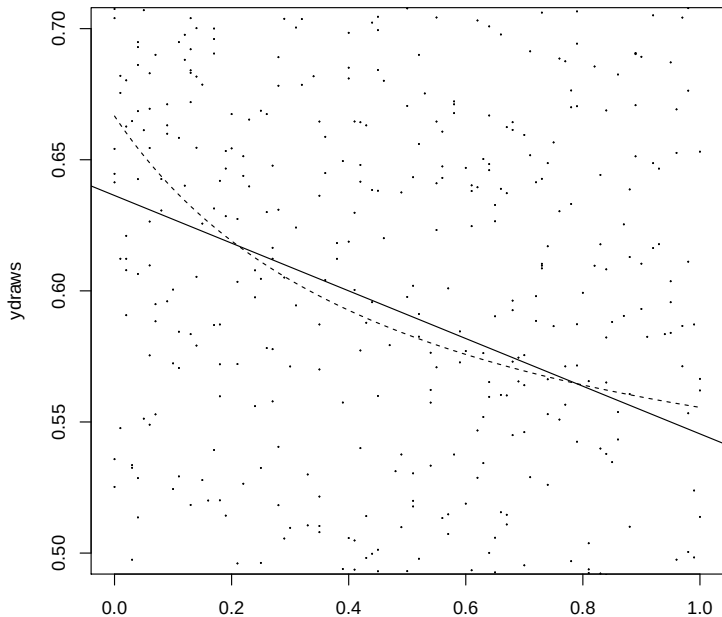
Example: $X + Y$

Suppose X and Y have pdf $x + y$ for $x, y \in [0, 1]$.

$$b = \frac{\text{cov}(X, Y)}{\text{Var}(X)} = -1/11$$

$$a = E(Y) - bE(X) = 84/132$$





Sums of Random Variables

Suppose we have a sequence of random variables X_i , $i = 1, 2, \dots, N$.
Suppose that they have joint pdf,

$$f(\mathbf{x}) = f(x_1, x_2, \dots, x_n)$$

- 1) $E[\sum_{i=1}^N X_i] = \sum_{i=1}^N E[X_i]$
- 2) $\text{var}(\sum_{i=1}^N X_i) = \sum_{i=1}^N \text{var}(X_i) + 2 \sum_{i < j} \text{cov}(X_i, X_j)$

Sums of Random Variable

Proof.

Consider two random variables, X_1 and X_2 . Then,

$$\begin{aligned}\text{var}(X_1 + X_2) &= E[(X_1 + X_2)^2] - (E[X_1] + E[X_2])^2 \\&= E[X_1^2] + 2E[X_1X_2] + E[X_2^2] \\&\quad - (E[X_1])^2 - 2E[X_1]E[X_2] - E[X_2]^2 \\&= \underbrace{E[X_1^2] - (E[X_1])^2}_{\text{var}(X_1)} + \underbrace{E[X_2^2] - E[X_2]^2}_{\text{var}(X_2)} \\&\quad + 2 \underbrace{(E[X_1X_2] - E[X_1]E[X_2])}_{\text{cov}(X_1, X_2)} \\&= \text{var}(X_1) + \text{var}(X_2) + 2\text{cov}(X_1, X_2)\end{aligned}$$



Definition

Suppose $\mathbf{X} = (X_1, X_2, \dots, X_N)$ is a vector of random variables. If $\mathbf{X}$ has pdf

$$f(\mathbf{x}) = (2\pi)^{-N/2} \det(\mathbf{\Sigma})^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \mathbf{\Sigma}(\mathbf{x} - \boldsymbol{\mu})\right)$$

Then we will say $\mathbf{X}$ is a **Multivariate Normal** Distribution,

$$\mathbf{X} \sim \text{Multivariate Normal}(\boldsymbol{\mu}, \mathbf{\Sigma})$$

- **Regularly** used for likelihood, Bayesian, and other parametric inferences

Multivariate Normal Distribution

Consider the (bivariate) special case where $\boldsymbol{\mu} = (0, 0)$ and

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Then

$$\begin{aligned} f(x_1, x_2) &= (2\pi)^{-2/2} 1^{-1/2} \exp \left(-\frac{1}{2} \left((\mathbf{x} - \mathbf{0})' \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} (\mathbf{x} - \mathbf{0}) \right) \right) \\ &= \frac{1}{2\pi} \exp \left(-\frac{1}{2} (x_1^2 + x_2^2) \right) \\ &= \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{x_1^2}{2} \right) \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{x_2^2}{2} \right) \end{aligned}$$

$\rightsquigarrow$ product of univariate standard normally distributed random variables

Standard Multivariate Normal

Definition

Suppose $\mathbf{Z} = (Z_1, Z_2, \dots, Z_N)$ is

$$\mathbf{Z} \sim \text{Multivariate Normal}(\mathbf{0}, \mathbf{I}_N).$$

Then we will call $\mathbf{Z}$ the standard multivariate normal.

Properties of the Multivariate Normal Distribution

Suppose $\mathbf{X} = (X_1, X_2, \dots, X_N)$

$$\begin{aligned} E[\mathbf{X}] &= \boldsymbol{\mu} \\ \text{cov}(\mathbf{X}) &= \boldsymbol{\Sigma} \end{aligned}$$

So that,

$$\boldsymbol{\Sigma} = \begin{pmatrix} \text{var}(X_1) & \text{cov}(X_1, X_2) & \dots & \text{cov}(X_1, X_N) \\ \text{cov}(X_2, X_1) & \text{var}(X_2) & \dots & \text{cov}(X_2, X_N) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(X_N, X_1) & \text{cov}(X_N, X_2) & \dots & \text{var}(X_N) \end{pmatrix}$$

Independence and Multivariate Normal

Proposition

Suppose X and Y are independent. Then

$$\text{cov}(X, Y) = 0$$

Proof.

Suppose X and Y are independent.

$$\text{cov}(X, Y) = E[XY] - E[X]E[Y]$$

Calculating $E[XY]$

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf_X(x)f_Y(y) dx dy \\ &= \int_{-\infty}^{\infty} xf_X(x) dx \int_{-\infty}^{\infty} yf_Y(y) dy \\ &= E[X]E[Y] \end{aligned}$$

Then $\text{cov}(X, Y) = 0$.



Zero covariance does not **generally** imply Independent

Suppose $X \in \{-1, 1\}$ with $P(X = 1) = P(X = -1) = 1/2$.

Suppose $Y \in \{-1, 0, 1\}$ with $Y = 0$ if $X = -1$ and $P(Y = 1) = P(Y = -1)$ if $X = 1$.

$$\begin{aligned}E[XY] &= \sum_{i \in \{-1, 1\}} \sum_{j \in \{-1, 0, 1\}} ijP(X = i, Y = j) \\&= -1 \times 0 \times P(X = -1, Y = 0) + 1 \times 1 \times P(X = 1, Y = 1) \\&\quad -1 \times 1 \times P(X = 1, Y = -1) \\&= 0 + P(X = 1, Y = 1) - P(X = 1, Y = -1) \\&= 0.25 - 0.25 = 0 \\E[X] &= 0 \\E[Y] &= 0\end{aligned}$$

Proposition

*Suppose $\mathbf{X} \sim \text{Multivariate Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. where $\mathbf{X} = (X_1, X_2, \dots, X_N)$.
If $\text{cov}(X_i, X_j) = 0$, then X_i and X_j are independent*

Iterated Expectations

Proposition

Suppose X and Y are random variables. Then

$$E[X] = E[E[X|Y]]$$

- Inner Expectation is $E[X|Y] = \int_{-\infty}^{\infty} xf_{X|Y}(x|y)dx$.
- Outer expectation is over y .

Iterated Expectations

Proof.

$$\begin{aligned}E[E[X|Y]] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X|Y}(x|y) f_Y(y) dx dy \\&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X|Y}(x|y) f_Y(y) dy dx \\&= \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f(x, y) dy dx \\&= \int_{-\infty}^{\infty} x f_X(x) dx \\&= E[X]\end{aligned}$$



Iterated Expectations

Definition

Suppose Y is a continuous random variable with $Y \in [0, 1]$ and pdf of Y given by

$$f(y) = \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y^{\alpha_1-1} (1-y)^{\alpha_2-1}$$

Then we will say Y is a **Beta** distribution with parameters α_1 and α_2 . Equivalently,

$$Y \sim \text{Beta}(\alpha_1, \alpha_2)$$

- Beta is a distribution on **proportions**
- Beta is a special case of the **Dirichlet** distribution
- $E[Y] = \frac{\alpha_1}{\alpha_1 + \alpha_2}$

Iterated Expectations

Suppose

$$\begin{aligned}\pi &\sim \text{Beta}(\alpha_1, \alpha_2) \\ Y|\pi, n &\sim \text{Binomial}(n, \pi)\end{aligned}$$

What is $E[Y]$?

$$\begin{aligned}E[Y] &= E[E[Y|\pi]] \\ &= \int_{-\infty}^{\infty} \sum_{j=0}^N \binom{N}{j} j p(j|\pi) f(\pi) d\pi \\ &= \int_{-\infty}^{\infty} N\pi f(\pi) d\pi \\ &= N \frac{\alpha_1}{\alpha_1 + \alpha_2}\end{aligned}$$

Conditional Expectation as an Estimator

Suppose Y is a random variable, and X is some data.

$$\hat{Y} = E[Y|X]$$

So we know $\hat{Y}$ is an estimator.

Estimation error

Suppose $\hat{Y}$ is an estimator from before, the estimation error:

$$\tilde{Y} = \hat{Y} - Y$$

is a random variable where:

$$E[\tilde{Y}|X] = E[\hat{Y} - Y|X] = E[\hat{Y}|X] - E[Y|X] = \hat{Y} - \hat{Y} = 0$$

By iterated expectations:

$$E[\tilde{Y}] = E[E[\tilde{Y}|X]] = 0$$

This estimator's error centers on 0.

Correlation of Estimator and Estimation Error

Suppose $\hat{Y}$ is an estimator from before, and $\tilde{Y}$ the estimation error.

$$E[\tilde{Y}\hat{Y}] = E[E[\tilde{Y}\hat{Y}|X]] = E[\hat{Y}E[\tilde{Y}|X]] = 0$$

because $\hat{Y}$ only depends on X .

$$\text{cov}(\tilde{Y}, \hat{Y}) = E[\tilde{Y}\hat{Y}] - E[\tilde{Y}]E[\hat{Y}] = 0 - E[\tilde{Y}]0 = 0$$

The error is uncorrelated with the estimate. This is what allows us to separate the signal from the noise.

Law of Total Variance

Suppose X and Y are random variables. Then

$$\text{var}[X] = E[\text{var}[X|Y]] + \text{var}(E[X|Y])$$

The first term is the average of the variation for each value of Y . The second term is the variation across the averages within each value of Y .

Law of Total Variance: application

Suppose θ is a parameter, and x is data.

$$\text{var}[\theta] = E[\text{var}[\theta|x]] + \text{var}(E[\theta|x])$$

The expected variance of θ conditional on data (the posterior) is less than the variance in θ without the data (the prior).